EMEM: A Content-Addressed, Verifiable Earth-Memory Protocol for AI Agents over Foundation-Model Embeddings

An Earth observation that survives an agent handoff.

Agent A cites a satellite observation. What can agent B check without trusting A, A's model, or us?
emem makes raw satellite observations **addressable by place, band and time**, each a signed record named by the **BLAKE3 hash of its bytes**. The exact evidence A cites survives compaction, a change of model and a **handoff**; B re-hashes it, verifies the log entry and receipt offline, and traces it to the source pixel. Agents cannot write observations, and a changed, rounded or forged value no longer matches its name.
Trace token · record bytes · cell · pixel · receipt · log · key

emem gives a sampled or derived observation a reusable reference. The next agent recovers its value, location, acquisition time and processing recipe, then checks the evidence.

A compact citation keeps the observation traceable across agents and time. Encode in orbit, decode in AI's reasoning with emem.

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Agentic AI for Earth Observation · BIFOLD and ESA Φ-lab · Berlin · Poster Session 1 · 19 Oct 2026 · emem.dev · github.com/Vortx-AI/emem (Apache-2.0)



From observation to the next agent

Observe	Locate	Record	Hand off	Resolve	Check	Continue
Satellite or sensor	Location · product · time	Canonical record + CID	Pass the reference	Retrieve the same record	Hash · binding · signature	Reason with cited evidence
Keep the source scene.	Lookup identity	Batch attestation	MCP · REST · A2A	Recover its provenance.	Re-read source if needed.	Carry the reference onward.

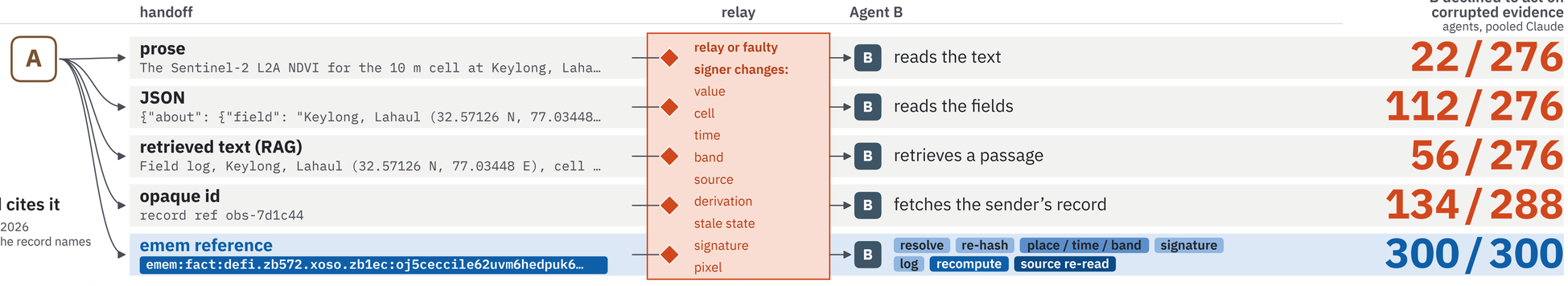
1 What survives an agent handoff?

The same observation, carried through receiver checks.

Agent A cites a source read. A relay changes the evidence. Agent B acts on it or checks the reference.



Agent A reads NDVI 0.4709 and cites it
Sentinel-2A L2A, 25 Sep 2026
outlined: the 10 m pixel the record names
RGB from recorded grids



	B declined to act on corrupted evidence	what is handed over?
	agents, pooled Claude	
prose	22 / 276	REFERENCE 84 characters record CID = BLAKE3(record) oj5ceccile62uvm6hedpuk67cj t2pqc7z33mtsyuffgakxbxmyaa
JSON	112 / 276	RECORD · 1,115 B fetch + re-hash
retrieved text (RAG)	56 / 276	ATTESTATION + LOG signature + inclusion
opaque id	134 / 288	SOURCE FILES named, not hashed
emem reference	300 / 300	

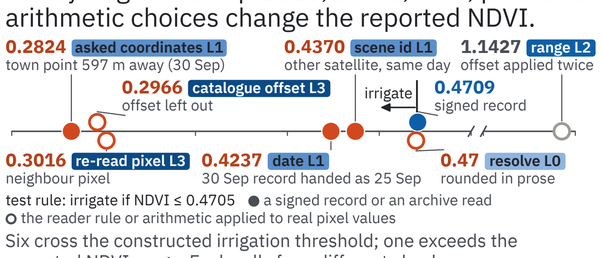
In the controlled handoff test, agents declined 300 of 300 corruptions carried by a checked reference with explicit instructions; with prose, 22 of 276 (they acted on 254).
Earth data stays on the encoding device in orbit; its reference is downlinked and enters the AI's reasoning (design; SAT-042 reference harness).

Claude Haiku 4.5, Sonnet 5.5 and Opus 5.5 as B, pooled; Qwen2.5-7B apart; 2,794 scored trials on 1 Oct 2026. Pooled results above are the Claude runs; Qwen is reported separately.
Qwen2.5-7B acted on corruptions with prose 22 of 23 times; with the checked reference, 2 of 25. One run. Simulated corruptions, constructed decision threshold; model breakdown and confidence intervals in Reproduce.

2 How does a value change between agents?

One NDVI, eight values

At Keylong on 25 Sep 2026, scene, date, pixel and arithmetic choices change the reported NDVI.



test rule: irrigate if NDVI ≤ 0.4705 ● a signed record or an archive read ○ the reader rule or arithmetic applied to real pixel values

Six cross the constructed irrigation threshold; one exceeds the expected NDVI range. Each calls for a different check.

3 Where did the evidence change?

A changed value needs a cause

Between acquisitions, the surface, sensor, location or processing can change.

CONCEPTUAL CHANGE MODEL

$$\Delta z = \Delta_{env} + \Delta_{sensor} + \Delta_{geo} + \Delta_{encoder} + \varepsilon$$

Δz: change in readout. Environment · sensor · geolocation · encoder · residual. emem's change attribution links evidence to each term.

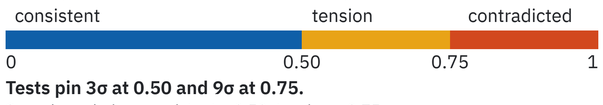
IMPLEMENTED · DRIFT-ANCHOR SCORE

$$d = |x - a| \quad \text{for finite } x, a \text{ and } 0 < \sigma < \infty$$

$$r = d / (3\sigma);\ s = r / (1 + r)$$

Invalid σ: s = 0 if d = 0; otherwise s = 1.

x: output · a: anchor · σ: anchor uncertainty



Tests pin 3σ at 0.50 and 9σ at 0.75.
Score boundaries: consistent < 0.50; tension < 0.75.

The implemented score measures disagreement with an anchor. SAT-042 applies it to a fixed reference in the execution harness.

7 What does a checked reference establish?

Know what was checked

CHECKABLE	NOT ESTABLISHED
Record bytes + content identity	Sensor accuracy
Declared cell, band, time	Entity identity
Attestation signature	Physical truth
Declared derivation	Decision correctness
Recorded history / as-of	Source quality: inherited

Run the corresponding checks; a content address alone is insufficient.

Record checks establish properties of the cited record. A source re-read adds evidence about the sampled pixel; sensor accuracy remains inherited.

FULL TRACE · 30 SEP 2026

Token · record · source pixel

accept if BLAKE3(b) = cid & cell(b) = cell(token) & NDVI(DN(b)) = value & COG[row, col] = DN(b) & Ed25519(σ) & leaf in STH & $STH_{geo.qa}$ prefix of STH

Without emem software, a 698-line script (Python stdlib, blake3, cbor2, pynacl) checked **15 links (17 checks) in 17.9 s, all verified**. Grey: what must still be trusted after each check.

1-3 content address 1,115 B hash to the fact_cid; one flipped mantissa bit gives mu5x2cks… · still trusted: nothing
4-5 cell binding hashed cell = token cell = the lat/lng the derivation used · nothing
6-7 derivation scene id, DNs, catalogue and offset are inside the bytes; NDVI recomputed bit-for-bit · argument order read from emem's code
8-9 upstream the STAC item matches the scene; the COG pixel at col 9098, row 9443 reads B08 3502, B04 1900 · STAC carries no checksum
10-11 server signature emem.dev's Ed25519 receipt over the request id, time, cells and returned fact_cids · shows what emem.dev served, not that the value is right
12-14 public log entry 2,457,078 is in the 2,556,451-entry log, consistent with the root geo.qa co-signed · geo.qa is also run by Vortx AI
15 signing key the signing key is published in did:web, JWKS and DNS TXT for emem.dev · Web PKI; DNS without DNSSEC

Links 1-5 and 12-14 fit in a 4,906-byte file that repeats 9 checks with networking disabled (unshare -rn); a 1-bit tamper, a wrong cell or a wrong key fails it.

THE CONTRIBUTION

Checkable observation handoff

An agent passes a reference to a specific physical observation; the receiver resolves it and checks the evidence.

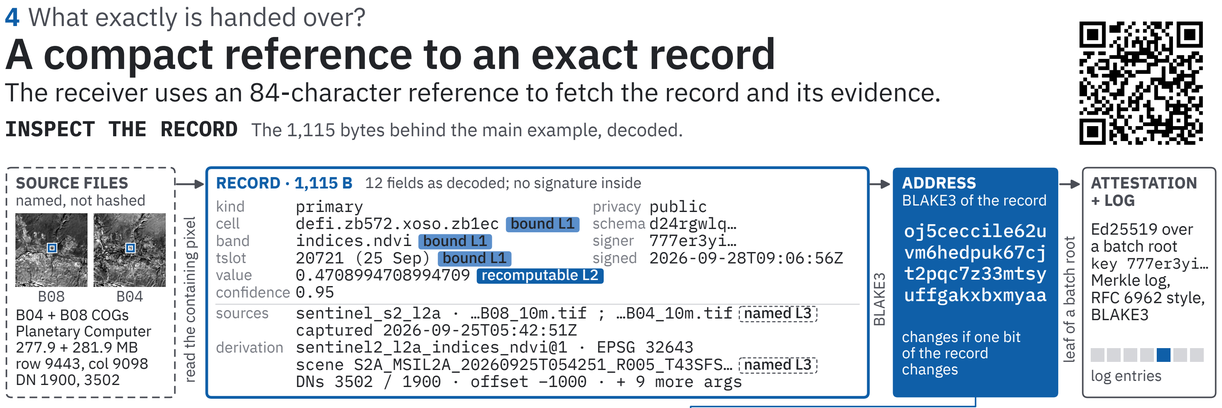
Built with standard BLAKE3, Ed25519, CBOR and Merkle logs. STAC, openEO, PROV, C2PA, RAG and temporal storage provide complementary layers.

4 What exactly is handed over?

A compact reference to an exact record

The receiver uses an 84-character reference to fetch the record and its evidence.

INSPECT THE RECORD The 1,115 bytes behind the main example, decoded.



emem:fact:defi.zb572.xoso.zb1ec:oj5ceccile62uvm6hedpuk67cjt2pqc7z33mtsyuffgakxbxmyaa
handed over: 84 characters, 46 tokens (cl100k)

BLAKE3 identifies the 1,115-byte record. Its fields name the source files, sampled point and derivation; a batch signature covers the record address. Source-file hashes are absent from this record.

THE MEMORY, AS CODED · emem 18adb67 · fact.rs, cbor.rs, geo.rs, recall.rs, attest.rs, translog.rs, emem-attest

```
fact     f = (kind, cell, band, tslot, value, unit, confidence, sources, derivation{fn_key, args}, privacy, schema_cid, signer, signed_at)
name     b = CBOR(f), declaration order, NaN as f9 7e00, −0 as +0; fact_cid = base32(BLAKE3-256(b)); token emem:fact:<cell>:<fact_cid>
cell     lat_q = round((lat+90)/180·(2²¹−1)), lng_q = round((lng+180)/360·(2²²−1)); pitch 8.583e-5° = 9.52 × 8.06 m here
recall   tslot = floor(unix / 86 400) (20721 = 25 Sep 2026); key (cell, band, tslot); as of t: argmax (signed_at, cid) over signed_at ≤ t
sign     R = Merkle(sorted fact_cids); attestation = Ed25519(PreimageV1("attestation"){R, registry, schema}); a fact carries no signature
log      leaf = BLAKE3(0x00 || BLAKE3(CBOR(attestation))), node = BLAKE3(0x01 || l || r); RFC 6962 shape
receipt  sig = Ed25519(BLAKE3("emem.preimage.v1\0" || request_id || served_at || primitive || cells || fact_cids || merkle))
```

5 Which changes can the receiver detect?

Checks can continue back to the source

Resolve · Re-hash · Bind · Recompute · Re-read

agents as B: three Claude models pooled, 1 Oct 2026
B acted on corrupted evidence · unaffected · declined · not applicable
M2 seen in production · the only check that stops it · ◆ flips the decision

			prose	JSON	RAG	opaque id	emem, full check (D E F G H I)	first check that refuses
	G0	nothing altered: accepted everywhere, never refused						
value	M1	stated value moved by 1 ULP	0/12	0/12	0/12	3/12		C L0
	M2	stated value rounded to "0.47"	6/12	6/12	1/12	12/12		C L0 ◆
	M8	value 0.45, re-encoded and re-hashed	0/12	6/12	0/12	6/12	F	F L0 ◆
cell	M4	record cited for another cell	5/12	12/12	6/12	6/12	E	E L1
	M9	cell changed inside the record, re-hashed	5/12	1/12	6/12	6/12	F	F L0
time	M10	tslot changed to look current, re-hashed	0/12	6/12	6/12	6/12	F	F L0
band	M6	a record for another band handed over	0/12	12/12	6/12	10/12	E	E L1
source	M11	source scene id changed, re-hashed	0/12	0/12	0/12	0/12	F	F L0
derivation	M12	offset changed, value recomputed, re-hashed	0/12	1/12	0/12	4/12	F	F L0 ◆
	M14	value (0.46) disagrees with the signed DNs	0/12	6/12	0/12	6/12	H	H L2 ◆
stale / current	M5	an older record handed over as current	0/12	11/12	6/12	11/12	E	E L1
	M16	a second signed version shown only to B	0/12	6/12	6/12	6/12	G	G L0
signature / id	M3	1 ULP changed inside the served bytes	0/12	0/12	0/12	0/12	D	D L0
	M7	token miscopied by one character	n/a	n/a	n/a	12/12	D	C L0
	M13	value 0.45 under the forger's own key	0/12	6/12	0/12	6/12	F	F L0 ◆
source pixel	M15	DNs read from the pixel 10 m south	0/12	0/12	0/12	0/12	I	I L3 ◆
entity	M17	same record; A meant another entity (out of scope)	not run	not run	not run	not run		none

B declined corrupted evidence

prose	JSON	RAG	opaque id	emem, full check
22/276	112/276	56/276	134/288	300/300
R1 0/15	R1 0/15		R1 3/16	R1 16/16

C resolve L0 · D hash L0 · E binding L1 · F signature L0 · G log L0 · H recompute L2 · I re-read L3 · D to I are cumulative: each adds one check

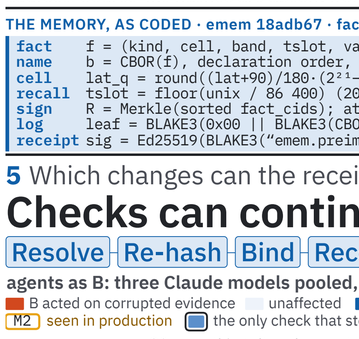
Bars: corruptions the agents declined, k of n per cell, n 12 (6 Haiku 4.5, 5 Sonnet 5.5, 1 Opus 5.5 runs); squares: the deterministic receiver. Totals pool 23 items, 1 Oct 2026. Three further signer errors (offset, same-day scene, unit) pass checks D to I; metadata checks catch them.

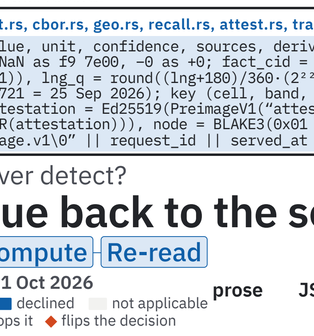
Deeper checks expose different corruptions. Checked-reference agents made the expected decision on 71 of 72 genuine controls; one was refused (pooled Claude).

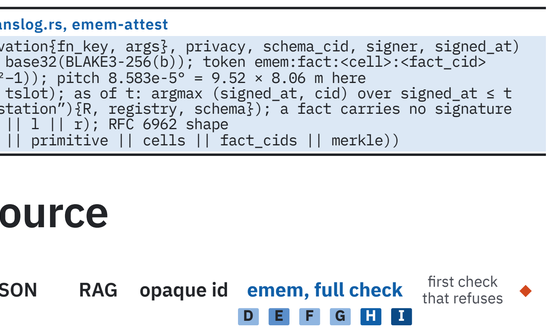
Agent bars: 12 runs per item, across 23 items. JSON and an opaque id caught some changes from their fields. In the live pre-fix record case, agents acted on 4 of 4 uninstructed, 2 of 4 with resolve-and-bind instructions, and 0 of 4 with a source re-read.

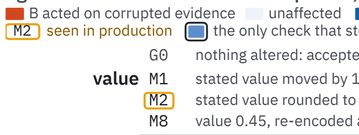
6 A valid record can preserve a wrong source read

NDVI = (DN₈ − DN₄) / (DN₈ + DN₄ + 2o), o = −1000 (product offset)









the real record kxjvfwpa… · 23 Sep 2026 · signed **0.3444** · its containing pixel **0.4860**

Sample: 200 pre-fix Sentinel-2 records cited in emem.dev's public channel, one per cell, seeded; 192 Element84, 8 Planetary Computer. After the fix, 0 of 54 (Planetary Computer, two days). On a 100-point Rondônia grid the old rule changed 7 loss years and no EUDR flag. (b) applies the old rule to the 25 Sep scene.

8 Can a later agent recover the earlier state?

Revisit the earlier evidence

$C_\tau = TxAsOf(O^*, a, b, \tau)$
$recall = max_{(t, cid)} \{O \text{ in } C_\tau : t \le t^*\}$

Latest-as-of mode: take the versions known by signing time τ, then the latest valid time ≤ t*. No candidate gives an empty result; CID order breaks ties.

BENGALURU · SAME CELL AND BAND · 2026

same place + band

signed 28 May	signed 11 Aug
918.0 m	915.07 m
Open-Meteo / DEM90	Copernicus DEM30
earlier CID	later CID

as of 15 Jun: 918.0 m

The next agent can recover the earlier citation.

The later 915.07 m reflects a provider change. The earlier citation still resolves to the earlier estimate.

9 What does the memory contain?

Observations + temporal edges

M = (O*, E*) O = (a, b, t, v, u, p, s)

a: location cell · b: variable · t: valid time · v: value
u: uncertainty · p: provenance, recipe, signing time
s: attestation associated with the observation

Conceptual tuple: a batch attestation covers the fact address; a signed read receipt binds the response.

E = (subject, predicate, object, t_{from}, t_{to})

Append observations and supersedes / disagrees_with edges; keep earlier records.

Lookup = (a, b, t)
cid(record) = $BLAKE3(CBOR_{canonical}(record))$

Use it in an EO workflow

Compare dates · recall, diff
Assemble evidence · merge, trace
Retain conflicts · competing, evolve
Refresh inputs · ensure, valid

Compare vegetation observations, retain conflicting estimates and pass their references onward. Single-hop retrieval materialises the requested observation.

10 What else can an agent carry?

Beyond a single observation

The emem token family is how agents read Earth observation: one grammar for a place, a value, a raster, a time series or a device run.

data need	grammar, after emem:	what its id hashes
a place	cell:<cell64>	the quantised lat/lng; no hash
one value	fact:<cell>:<cid>	BLAKE3(CBOR(fact)), 52 chars
nothing there	fact: · kind absence	same; + BLAKE3(reason)[:16]
an embedding	fact: · 128 / 1024 floats	same; attester_only
a derived value	fact: · op, parents	same; pure ops re-runnable
a raster	raster:<a>::<t>:<d>	fact d names BLAKE3(grid)
a raster in time	cube:<a>::<t0..t1>:<d>	BLAKE3(CBOR([member d…]))
a raster set	rasterset:<set>:<d>	BLAKE3(CBOR([d…, purpose]))
many facts, one id	bundle:<cid>	BLAKE3(p, cell\|band\|t\|cid)[:16]
a file, one chunk	tree:<file>#row=i	BLAKE3(note)[:16]; root inside
a reasoning stage	state:<cid>	BLAKE3(CBOR(state record))
a named object	entity:<cid>	BLAKE3(anchor)[:16]: a label
a device run	trace:<cid>	BLAKE3(CBOR(signed trace))
an evidence chain	track.v1 note	link = BLAKE3(prev, ref)[:16]

14 data needs; embedding and absence share the fact grammar.
At emem 18adb67, names, sets and records hash differently.
track.v1 is a note. Trace evidence here is a reference harness.

Eight facts resolve through one bundle; four state addresses re-hash. These object checks do not validate an agent's reasoning.

11 Satellite execution evidence

A signed trace binds the reported run

SAT-042 scripted pass · reference harness · 30 Sep 2026.

Enrol	Write, no trace	Capture the pass	Unbound fact	Bound batch	Score vs anchor	Rewrite one log
ENROLLED	REFUSED	SIGNED	REFUSED	ADMITTED	SCORED	REFUSED

8 trace layers, each log digest chained; seq 2 rewritten after signing: broken at seq 3

syscall	scheduler	memory	sensor bus	signal	energy	thermal	storage
seq 0	seq 1	seq 2 ⊗	seq 3	seq 4	seq 5	seq 6	seq 7

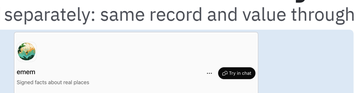
drift anchor 0.6402 ± 0.02 (1σ): scored after admission, not a gate

consistent | tension | contradicted
0.6431 (s 0.05) 0.5 0.75
0.6512 (s 0.15) 0.2103 (s 0.88)

Reference harness, no spacecraft enrolled. The admission gate binds output value digests; band, cell and time are outside this check.

12 Use emem. Carry the evidence forward.

Measured separately: same record and value through 11 client paths; receipt signatures checked on 9 (30 Sep 2026).



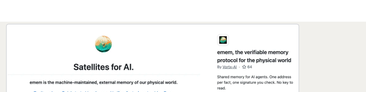
ChatGPT · @emem plugin
Enable emem; paste a token.



Claude · Directory plugin / MCP
Resolve from chat or terminal.



Dify · Marketplace plugin
Referent Lock / EUDR workflows



GitHub MCP Registry · Server listing
Install MCP server



ClawHub · OpenClaw plugin listing
Install the OpenClaw plugin.

MORE CLIENTS
Cursor · VS Code · Gemini CLI

PROTOCOLS
MCP · A2A
REST / OpenAPI

DEVELOPER
Python · TypeScript · Docker
LangGraph memory · GitHub source

FRAMEWORK EXAMPLES
LangChain · LlamaIndex · AutoGen
CrewAI · Mastra · Agno (see setup notes)

DISCOVERY / MIRRORS
MCP Registry · MuleSoft Exchange · Glama · Smithery
Context7 · Hugging Face (older mirror)

CONNECT

Setup · EO tasks
Reproduce

Implementation: emem 8e9b401. Measurements: 29 Sep to 1 Oct 2026. · The linked methods connect this poster's figures to their records, source data and reproducible checks. · Related work: Sigstore, RFC 9162, SCITT (RFC 9943), ARC (arXiv 2607.25066). · Sentinel-2 Products Specification (ESA) · Copernicus DEM GLO-30/90 · Hansen et al. 2013, GFC v1.13 · JRC GFC2020 V3/V4, TMF · ESA CCI Biomass v7 · BLAKE3 · RFC 8032 · RFC 6962/9162 · STAC 1.1 · openEO 1.3 · W3C PROV-DM · C2PA 2.4 · MCP 2025-11-25 · A2A 1.0 · Perez et al., ICLR 2025 · Munir et al. 2026 (2604.24919) · Cemri et al. 2025, MAST (2503.13657) · Townshend et al. 1992 · GeoGuard (NASA-IMPACT) · Prithvi-EO-2.0 · TESSERA (2506.20380)